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Repository: <https://github.com/Daazee/fraud-detection-models>

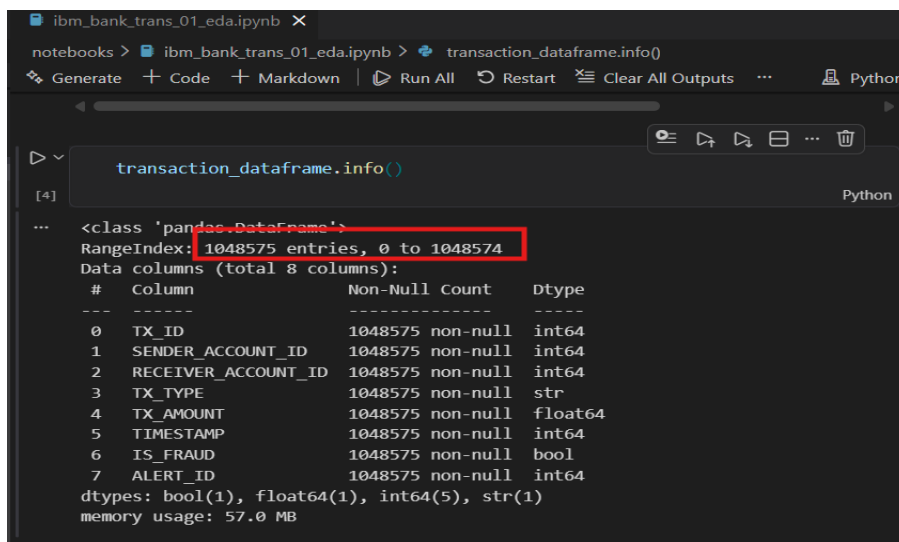
1.0 Completed Work

This report documents progress on the project titled "**An Explainable Machine Learning-Based Framework for Fraud Detection in Digital Banking Transfers.**" The literature review of eight peer-reviewed studies, scheduled for Weeks 1-4, has been completed. The project uses a publicly available, anonymised dataset and involves no human subjects or primary data collection, so formal ethics approval is not required. All code is version-controlled in the project repository.

1.1 Data Collection and Understanding

The EPP identified the IEEE-CIS Fraud Detection Dataset as the primary source. During exploratory data analysis (EDA), this dataset was found to comprise card-based payment transactions rather than inter-account banking transfers. It was therefore structurally unsuitable for the project scope. Alternative datasets were therefore researched. The IBM AMLSim Example Dataset was selected over PaySim, whose simulated mobile-money structure offers limited counterparty detail, because it preserves sender–receiver relationships and suspicious-activity flags, enabling the relational feature engineering and supervised–anomaly comparison this project requires. This relational structure aligns the dataset with the money-laundering detection setting (Irshad et al., 2024; Ketenci et al., 2021). The trade-off is that synthetic data does not capture production noise, so results demonstrate framework viability rather than deployment-ready performance.

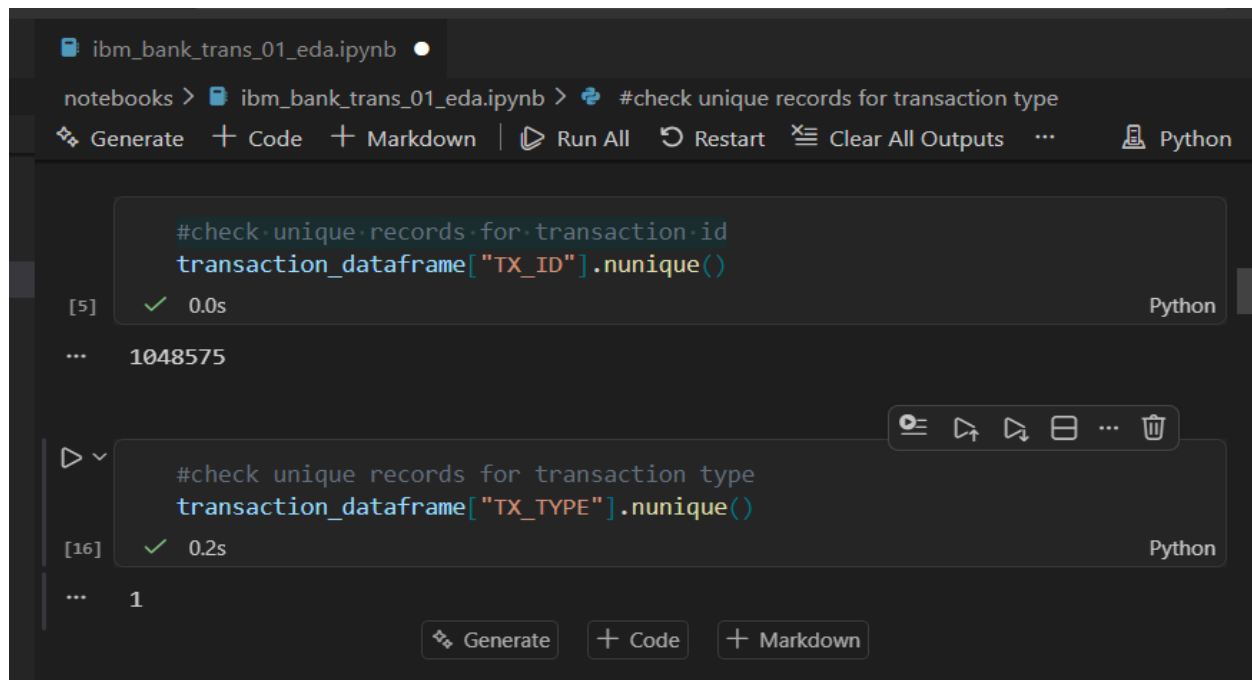
Of the dataset's files, only the transactions file (**transactions.csv**) is used for modelling; it is loaded through a configurable ingestion module (`src/ingestion/loaders.py`) and contains 1,048,575 records across eight columns with no missing values with no missing values. EDA confirmed severe class imbalance: only **1,329** transactions (**0.13%**) are fraudulent, motivating PR-AUC (Davis and Goadrich, 2006), as the primary metric and SMOTE during training.



```
transaction_dataframe.info()

<class 'pandas.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   TX_ID               1048575 non-null  int64  
 1   SENDER_ACCOUNT_ID   1048575 non-null  int64  
 2   RECEIVER_ACCOUNT_ID 1048575 non-null  int64  
 3   TX_TYPE              1048575 non-null  str      
 4   TX_AMOUNT            1048575 non-null  float64  
 5   TIMESTAMP            1048575 non-null  int64  
 6   IS_FRAUD             1048575 non-null  bool     
 7   ALERT_ID             1048575 non-null  int64  
dtypes: bool(1), float64(1), int64(5), str(1)
memory usage: 57.0 MB
```

Figure 1.1.1: Dataset overview containing 1,048,575 total records.



```
ibm_bank_trans_01_eda.ipynb
```

notebooks > ibm_bank_trans_01_eda.ipynb > #check unique records for transaction type

Generate + Code + Markdown | Run All Restart Clear All Outputs ... Python

```
#check unique records for transaction id
transaction_dataframe["TX_ID"].nunique()
```

[5] ✓ 0.0s Python

... 1048575

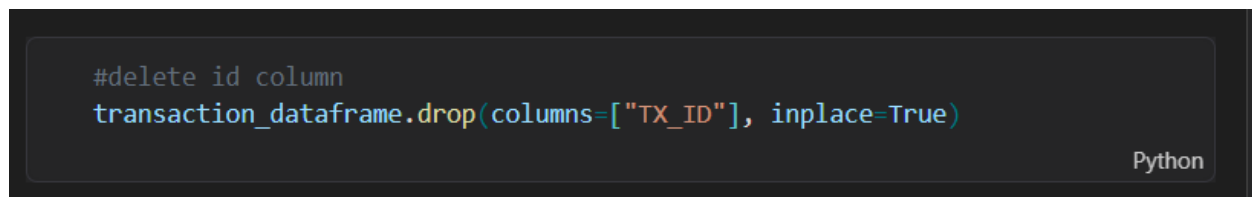
```
#check unique records for transaction type
transaction_dataframe["TX_TYPE"].nunique()
```

[16] ✓ 0.2s Python

... 1

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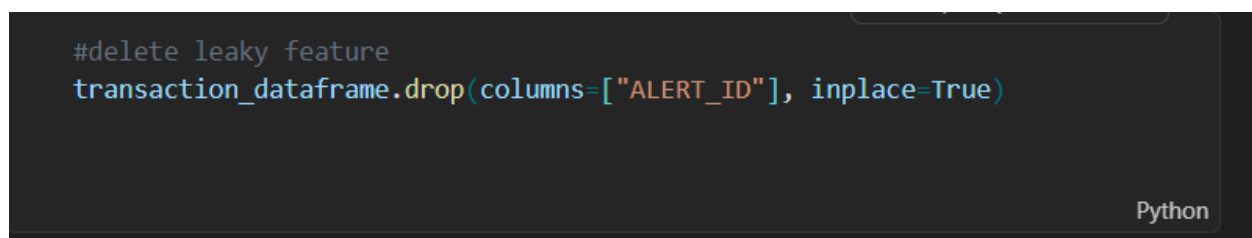
Figure 1.1.2: Evaluation of unique values per feature in the dataset.



```
#delete id column
transaction_dataframe.drop(columns=["TX_ID"], inplace=True)
```

Python

Figure 1.1.3: Deletion of dataset ID column



```
#delete leaky feature
transaction_dataframe.drop(columns=["ALERT_ID"], inplace=True)
```

Python

Figure 1.1.4: Deletion of dataset leaky feature

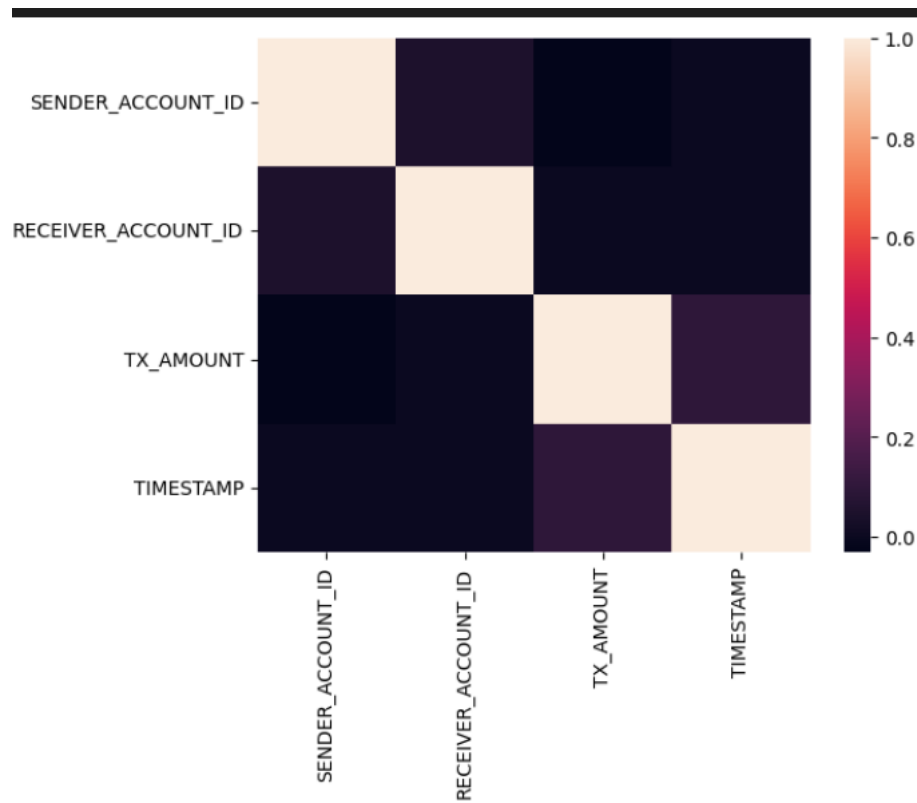


Figure 1.1.5: Heatmap indicating weak or no correlations among the features.

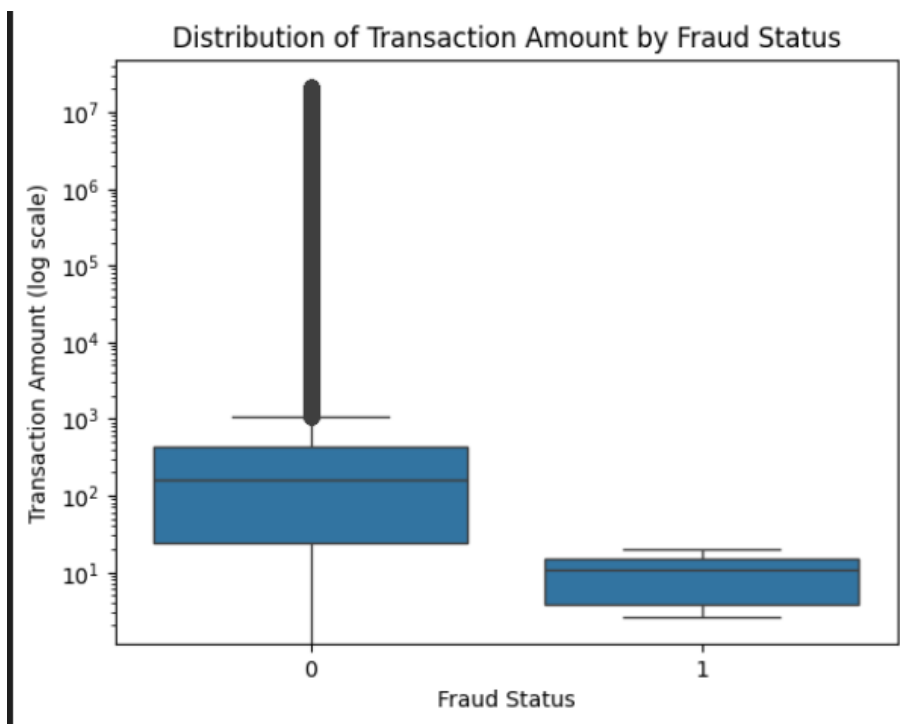
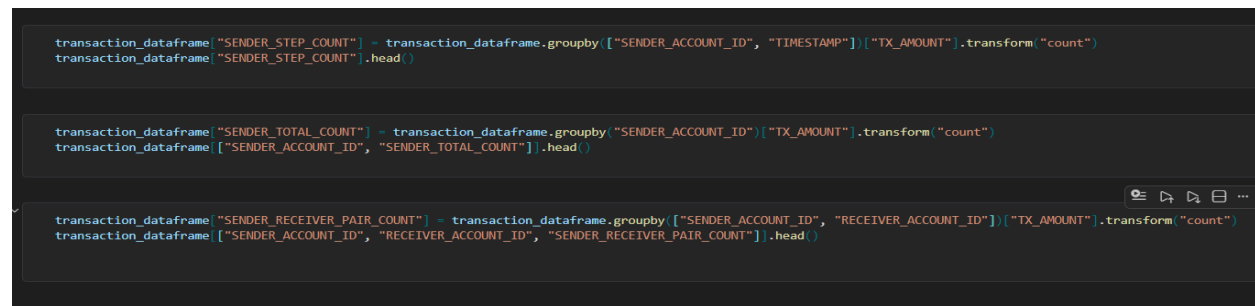


Figure 1.1.6: Boxplot showing the distribution of Transaction Amount by Fraud Status

1.2 Data Preprocessing

Three columns were removed: **TX_ID** (identifier), **ALERT_ID** (a post-hoc flag causing data leakage), and **TX_TYPE** (a single constant value). Pairwise correlation analysis dropped any feature exceeding 0.8 with weaker correlation to the target. A chi-squared test guided outlier treatment: features whose outliers were significantly fraud-associated were preserved, while the rest were winsorised at the 1st and 99th percentiles. Features with absolute skewness above 1.0, including **TX_AMOUNT** (skew 23.0), were log1p-transformed.

These choices reflect a deliberate priority: on severely imbalanced domains, careful data preparation governs detection quality more than accuracy-driven tuning (Branco et al., 2016). Raw features correlated only weakly with fraud, so, three behavioural velocity features were engineered: **SENDER_STEP_COUNT** (transactions per sender per time step, where exactly two corresponded to a 6.0% fraud rate), **SENDER_RECEIVER_PAIR_COUNT** (first-time pairs showed 8.2% fraud), and **SENDER_TOTAL_COUNT**. These raised XGBoost PR-AUC from 0.89 to 0.99.



```
transaction_dataframe["SENDER_STEP_COUNT"] = transaction_dataframe.groupby(["SENDER_ACCOUNT_ID", "TIMESTAMP"])["TX_AMOUNT"].transform("count")
transaction_dataframe["SENDER_STEP_COUNT"].head()

transaction_dataframe["SENDER_TOTAL_COUNT"] = transaction_dataframe.groupby("SENDER_ACCOUNT_ID")["TX_AMOUNT"].transform("count")
transaction_dataframe[["SENDER_ACCOUNT_ID", "SENDER_TOTAL_COUNT"]].head()

transaction_dataframe["SENDER_RECEIVER_PAIR_COUNT"] = transaction_dataframe.groupby(["SENDER_ACCOUNT_ID", "RECEIVER_ACCOUNT_ID"])["TX_AMOUNT"].transform("count")
transaction_dataframe[["SENDER_ACCOUNT_ID", "RECEIVER_ACCOUNT_ID", "SENDER_RECEIVER_PAIR_COUNT"]].head()
```

Figure 1.2.1: Engineered velocity features

1.3 Model Development

Four models were implemented. Each supervised model was encapsulated within an `imblearn.pipeline.Pipeline`, so SMOTE (Chawla et al., 2002) is fitted only on training data within each fit call and never applied to the test set. This prevents synthetic samples from leaking into evaluation and inflating performance. SMOTE was preferred over class weighting, which does not increase fraud-example diversity (the binding constraint at 0.13%), and over undersampling, which would discard most legitimate records. SMOTE known risk of noise near class overlaps is mitigated by confining it to each training fold, with class weighting retained as a fallback.

Logistic Regression used 1,000 maximum iterations to ensure convergence on the high-dimensional encoded feature space, with a `StandardScaler` step before the classifier, as it is sensitive to feature magnitude. **Random Forest** and **XGBoost** (Chen and Guestrin, 2016) were both parallelized.

Isolation Forest was trained on `X_train` without labels, isolating anomalies through random partitioning and allowing detection of unseen fraud patterns (Liu et al., 2008). Its contamination was set to **0.0013** to match the **0.13%** fraud rate, rather than the default **0.1**, which would flag roughly **77** times too many transactions and inflate the false positive rate (FPR).

```
(.venv) PS D:\My Stuff\MSc Study\Courses\Project\code\fr(Set-ExecutionPolicy -python src/main.py --preprocess
Running preprocessing...
Loaded: 1,048,575 rows, 7 features
Train: 838,860 | Test: 209,715
Fraud rate - train: 0.1267% test: 0.1268%

Training Logistic Regression...
Training Random Forest...
Training XGBoost...
Training Isolation Forest...
Training Dummy Baseline...

=====
```

Figure 1.2.2: Model training

1.4 Challenges Encountered

The initial dataset unsuitability was the most significant problem. It required restarting data preparation but ultimately strengthened the alignment between the data and the research questions. Class imbalance, which initially produced classifiers biased toward the majority class, was resolved through stratified splitting and training-only SMOTE (Chawla et al., 2002), with class-weighting retained as a fallback. Data leakage from ALERT_ID was identified and removed during EDA, and the Isolation Forest contamination default of 0.1 was corrected to 0.0013 to avoid overestimating anomalies. Further challenges included weak raw-feature correlations, which required engineering behavioural velocity features to expose fraud signal.

1.5 Initial Evaluation

Models were assessed on a held-out 20% test set (209,715 transactions) using Precision, Recall, F1-Score, ROC-AUC, PR-AUC, FPR, and FNR.

Table 1.5.1: Model performance on the held-out test set (209,715 transactions). Best values for the two-priority metrics, PR-AUC and FNR, are shown in bold.

	Precision	Recall	F1 Score	FPR	FNR	ROC-AUC	PR-AUC
Model							
Dummy (Baseline)	0.0000	0.0000	0.0000	0.0000	1.0000	0.5000	0.0013
Logistic Regression	0.0769	0.9323	0.1420	0.0142	0.0677	0.9859	0.2163
Random Forest	0.9924	0.9774	0.9848	0.0000	0.0226	0.9962	0.9921
XGBoost	0.9776	0.9850	0.9813	0.0000	0.0150	0.9991	0.9857
Isolation Forest	0.0264	0.0263	0.0264	0.0012	0.9737	0.8708	0.0074

```
— Model Comparison —
precision recall f1_score fpr fnr roc_auc pr_auc
model
Dummy (Baseline) 0.0000 0.0000 0.0000 0.0000 1.0000 0.5000 0.0013
Logistic Regression 0.0769 0.9323 0.1420 0.0142 0.0677 0.9859 0.2163
Random Forest 0.9924 0.9774 0.9848 0.0000 0.0226 0.9962 0.9921
XGBoost 0.9776 0.9850 0.9813 0.0000 0.0150 0.9991 0.9857
Isolation Forest 0.0264 0.0263 0.0264 0.0012 0.9737 0.8708 0.0074

— Best Model —
By PR-AUC : Random Forest (0.9921)
By F1-Score : Random Forest (0.9848)

Overall best (PR-AUC + F1): Random Forest
```

Figure 1.5.1: Evaluation pipeline output, confirming results were generated from the implemented code.

Table 1.5.1 shows all three supervised models exceeding the baseline PR-AUC of **0.0013** by a substantial margin, confirming effective fraud detection under severe class imbalance and addressing the first research question. Random Forest achieved the highest PR-AUC (**0.9921**) and the strongest combined PR-AUC and F1 rank, making it the overall best performer and the anchor for the hybrid stage. XGBoost was marginally behind on PR-AUC (**0.9857**) but recorded the lowest FNR (**0.0150**), missing the fewest fraudulent transactions; this narrow margin is why it is retained as a direct substitute should later threshold analysis favour minimising missed fraud. Logistic Regression achieved high recall (**0.9323**) but very low precision (**0.0769**), flagging many legitimate transactions as fraudulent and confirming its unsuitability as a standalone detector under this imbalance. Isolation Forest, operating without labels, returned low precision and a high FNR (**0.9737**) as expected for unsupervised detection, supporting its intended role as a complementary anomaly signal within the hybrid model rather than a standalone classifier.

[Section 1 Word Count: 914 words]

2.0 Work to be Done

The remaining work completes the hybrid model, final evaluation, and explainability, building directly on the pipeline already implemented.

2.1 Hybrid Model Development

A hybrid detector will combine Random Forest with Isolation Forest anomaly scores via weighted averaging of normalised probabilities and anomaly scores, with weights tuned on validation data. If this does not improve on the individual models, three fallbacks will be evaluated: threshold-based stacking, the anomaly score as a supervised feature, and a voting ensemble. The hybrid will be compared directly against its components, addressing the second research question.

2.2 Final Evaluation

The hybrid model will be evaluated on the held-out test set using the same seven metrics, with threshold analysis identifying the operating point that best balances fraud capture against false alarms

2.3 SHAP Explainability

SHAP (Lundberg and Lee, 2017), based on Shapley values from cooperative game theory, will be applied to the best-performing model using TreeExplainer for efficiency at scale. Both global feature importance and local explanations (waterfall and force plots) will be analysed on a stratified sample of roughly 5,000 test records. If the hybrid model is selected, SHAP will be applied to its supervised component with the anomaly score treated as an explanatory feature, addressing the third research question.

2.4 Outstanding Challenges

Two challenges remain. First, fusing an unsupervised anomaly score with calibrated supervised probabilities requires careful normalisation, as the outputs lie on different scales; weighted averaging with validation-tuned weights and the documented fallbacks mitigate this. Second, SHAP at one-million-row scale is computationally costly; sampling and TreeExplainer address this, with global feature importance retained as a minimum deliverable.

2.5 Write-up and Submission

The Final Project Report will document the complete pipeline, results, SHAP analysis, and a critical evaluation against the three research questions, accompanied by a demonstration video and a finalised repository. The work is feasible within the remaining timeline: each outstanding stage extends the implemented, version-controlled pipeline rather than requiring new infrastructure, the full tool chain is already integrated, and the documented fallbacks provide defined contingencies should the fusion or full local explanation prove prohibitive.

[Section 2 Word Count: 323 words]

3.0 References

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- Davis, J. and Goadrich, M. (2006) 'The relationship between precision-recall and ROC curves', *Proceedings of the 23rd International Conference on Machine Learning*, pp. 233-240. doi: 10.1145/1143844.1143874.
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